Linear model, ANOVA

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Outline

Definition

Let $\mathbf{Y} = (Y_1, .., Y_n)^T$ be a random (column) vector, $\mathbf{X}_{n \times k}$ given matrix of constants (design matrix - matice plánu) and $\boldsymbol{\beta} = (\beta_1, ..., \beta_k)^T$ vector of parameters. $\mathbf{Y}$ is governed by a linear model if:

1. $E[\mathbf{Y}] = \mathbf{X}\beta$;
2. $Var(\mathbf{Y}) = \mathbf{V}$ exists and is independent on β

It will be assumed further that $n > k$ (for simplicity).

Definition

We call $\mathbf{b}$ a linear estimator for the vector β if a matrix $\mathbf{U}_{k \times n}$ exists in a way that $\mathbf{b} = \mathbf{UY}$

Gauss-Markov theorem

Theorem

Let the rank of a matrix $\mathbf{X}$ be k and $\mathbf{V}$ regular matrix. Then efficient linear estimator $\mathbf{b}$ for β is:

$$\mathbf{b} = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{V}^{-1}\mathbf{Y}$$

with a variance-covariance matrix:

$$Var(\mathbf{b}) = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}$$

Example weighted arithmetic mean:
Assume that $Y_1, \ldots, Y_n$ are independent and $Y_i \sim N(\mu, \sigma_i^2)$, where σ_i are known. Find an estimator for μ

$Y_1; \dots; Y_n \sim N(\mu; \sigma_i^2)$; Y_i a Y_j jsou nezávislé

$$V^{-1}_{2\times 2} = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}^{-1} = \frac{1}{\sigma_1^2 \sigma_2^2} \cdot \begin{pmatrix} \sigma_2^2 & 0 \\ 0 & \sigma_1^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{1}{\sigma_2^2} \end{pmatrix}$$

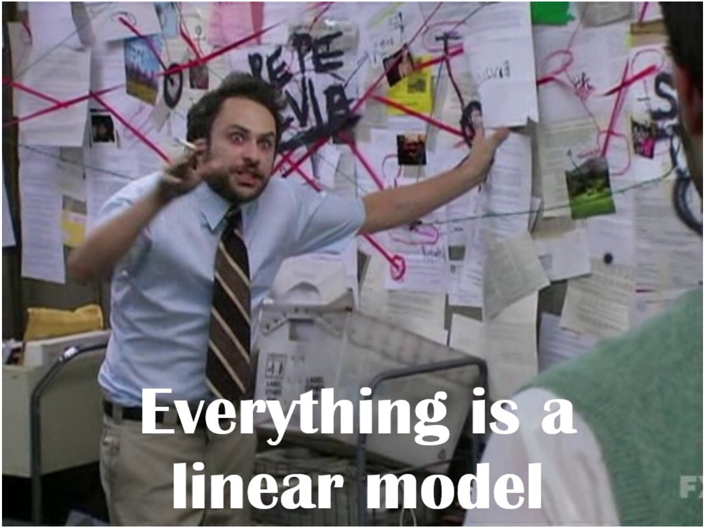
$$\mathbb{X}_{n\times 1} = \mathbb{1}_{n\times 1}$$

$$\mathbb{V}_{n\times n} = \operatorname{diag}\{\sigma_i^2\} \qquad \mathbb{V}^{-1}_{n\times n} = \operatorname{diag}\left\{\frac{1}{\sigma_i^2}\right\}$$

$$\hat{\mu} = b_{1\times 1} = (\mathbb{X}^T \mathbb{V}^{-1} \mathbb{X})^{-1} \mathbb{X}^T \mathbb{V}^{-1} \mathbb{Y} = \frac{1}{\sum_i \frac{1}{\sigma_i^2}} \cdot \mathbb{X}^T \mathbb{V}^{-1} \mathbb{Y} = \frac{1}{\sum_i \frac{1}{\sigma_i^2}} \cdot \sum_i \frac{Y_i}{\sigma_i^2}$$

$$\mathbb{X}^T \mathbb{V}^{-1} \mathbb{X} = \mathbb{1}^T_{n\times 1} \mathbb{V}^{-1}_{n\times n} \mathbb{1}_{n\times 1} = \left(\frac{1}{\sigma_1^2}; \dots; \frac{1}{\sigma_n^2}\right) \cdot \mathbb{1}_{n\times 1} = \sum_i \frac{1}{\sigma_i^2}$$

$$\mathbb{X}^T \mathbb{V}^{-1} \mathbb{Y} = \left(\frac{1}{\sigma_1^2}; \dots; \frac{1}{\sigma_n^2}\right) \mathbb{Y}_{n\times 1} = \sum_i \frac{Y_i}{\sigma_i^2}$$


Everything is a linear model

Theorem

Let $\mathbf{c} = (c_1, ..., c_k)^T$ be a nonzero vector, then $E[\mathbf{c}^T\mathbf{b}] = \mathbf{c}^T\boldsymbol{\beta}$ and $\hat{\theta} = \mathbf{c}^T\mathbf{b}$ is an estimator for $\theta = \mathbf{c}^T\boldsymbol{\beta}$ with variance $Var(\hat{\theta}) = \mathbf{c}^T(\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{c}$

Let's assume further that $\mathbf{V} = \sigma^2\mathbf{I}$.
Example: Using linear model and the assumption of constant (and known) variance, derive a test statistic for 2 normally distributed datasets testing $H_0 : \mu_1 = \mu_2$.

Definition

We call a vector $\mathbf{e} = (\mathbf{Y} - \mathbf{X}\mathbf{b})$ residual vector and $S_e = \mathbf{e}^T\mathbf{e}$ a residual sum of squares

Theorem

$s^2_{res} = \frac{S_e}{n-k}$ is an unbiased estimator of σ^2

$$\mathbb{Y}_{n\times 1} \sim N\left(\mathbb{X}_{n\times 2}\cdot \beta_{2\times 1};\ \sigma^2 \mathbb{I}_{n\times n}\right) \qquad H_0: \mu_1 = \mu_2 \qquad c^T\beta: (1;-1)\begin{pmatrix}\beta_1\\ \beta_1\end{pmatrix} \rightarrow H_0: \beta_1 - \beta_2 = 0$$

$n_1 + n_2 = n$

$$\mathbb{X}_{n\times 2} = \begin{pmatrix} \mathbb{1}_{n_1\times 1} & \mathbb{0}_{n_1\times 1} \\ \mathbb{0}_{n_2\times 1} & \mathbb{1}_{n_2\times 1}\end{pmatrix} \qquad (\mathbb{X}^T\cdot\mathbb{X})^{-1} = \begin{pmatrix} n_1 & 0 \\ 0 & n_2\end{pmatrix}^{-1} = \begin{pmatrix}\frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2}\end{pmatrix} \qquad \mathbb{X}^T\mathbb{Y} = \begin{pmatrix}\sum_{i=1}^{n_1} Y_i \\ \sum_{i=n_1+1}^{n} Y_i\end{pmatrix}$$

$$\mathbb{V}_{n\times n} = \sigma^2 \mathbb{I}_{n\times n} \rightarrow \underline{(\mathbb{X}^T\mathbb{V}^{-1}\mathbb{X})} \rightarrow \frac{1}{\sigma^2}(\mathbb{X}^T\mathbb{X}) \qquad \mathbb{b}_{2\times 1} = \cancel{\sigma^2}(\mathbb{X}^T\mathbb{X})^{-1}\cdot\frac{1}{\cancel{\sigma^2}}(\mathbb{X}^T\mathbb{Y})$$

$$\mathbb{b}_{2\times 1} = \begin{pmatrix}\frac{1}{n_1}\sum_{i=1}^{n_1} Y_i \\ \frac{1}{n_2}\sum_{i=n_1+1}^{n} Y_i\end{pmatrix} \qquad Var(\mathbb{b}_{2\times 1}) = \left(\frac{1}{\sigma^2}(\mathbb{X}^T\mathbb{X})\right)^{-1} = \sigma^2\cdot\begin{pmatrix}\frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2}\end{pmatrix} \qquad \mathbb{b}_{2\times 1} = \begin{pmatrix}\overline{Y}_1 \\ \overline{Y}_2\end{pmatrix}$$

$$E[c^T\mathbb{b}] = c^T\beta \qquad \hat{\theta} = c^T\mathbb{b} \qquad Var(\hat{\theta}) = c^T(\mathbb{X}^T\mathbb{V}^{-1}\mathbb{X})^{-1}c$$

$$\hat{\theta} = \overline{Y}_1 - \overline{Y}_2 \qquad Var(\hat{\theta}) = (1,-1)\begin{pmatrix}\frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2}\end{pmatrix}\begin{pmatrix}1\\-1\end{pmatrix}\cdot\sigma^2 = \sigma^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)$$

$$\frac{\overline{Y}_1 - \overline{Y}_2}{\sqrt{\sigma^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \sim N(0;1) \longrightarrow \frac{\overline{Y}_1 - \overline{Y}_2}{\sqrt{s^2_{res}\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \sim t(n-2)$$

Theorem

Let $\mathbf{c} = (c_1, ..., c_k)^T$ be a nonzero vector and $\mathbf{Y}_{n\times 1} \sim N(\mathbf{X}\beta; \sigma^2\mathbf{I})$ then

$$T = \frac{\mathbf{c}^T\mathbf{b} - \mathbf{c}^T\beta}{\sqrt{s_{res}^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}}} \sim t(n-k)$$

Example 1: Show that the test statistic for 1-dimensional t-test is consistent with preceding theorem
Example 2: Use previous results for comparing means of normally distributed data (for known constant variance) and relax the assumption of knowing the variance (it still needs to be the same)

$$\mathbb{Y}_{n\times 1} \sim N\left(\mathbb{X}_{n\times 2}\cdot \beta_{2\times 1};\ \sigma^2 \mathbb{I}_{n\times n}\right) \qquad H_0: \mu_1 = \mu_2 \qquad c^T\beta: (1;-1)\begin{pmatrix}\beta_1\\ \beta_1\end{pmatrix} \rightarrow H_0: \beta_1 - \beta_2 = 0$$

$$n_1 + n_2 = n$$

$$\mathbb{X}_{n\times 2} = \begin{pmatrix} \mathbb{1}_{n_1\times 1} & 0_{n_1\times 1} \\ 0_{n_2\times 1} & \mathbb{1}_{n_2\times 1}\end{pmatrix} \qquad (\mathbb{X}^T\cdot\mathbb{X})^{-1} = \begin{pmatrix} n_1 & 0 \\ 0 & n_2\end{pmatrix}^{-1} = \begin{pmatrix}\frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2}\end{pmatrix} \qquad \mathbb{X}^T\mathbb{Y} = \begin{pmatrix}\sum_{i=1}^{n_1} Y_i \\ \sum_{i=n_1+1}^{n} Y_i\end{pmatrix}$$

$$\mathbb{V}_{n\times n} = \sigma^2 \mathbb{I}_{n\times n} \rightarrow \underline{(\mathbb{X}^T\mathbb{V}^{-1}\mathbb{X})} \rightarrow \frac{1}{\sigma^2}(\mathbb{X}^T\mathbb{X}) \qquad \mathbb{b}_{2\times 1} = \cancel{\sigma^2}(\mathbb{X}^T\mathbb{X})^{-1}\cdot\frac{1}{\cancel{\sigma^2}}(\mathbb{X}^T\mathbb{Y})$$

$$\mathbb{b}_{2\times 1} = \begin{pmatrix}\frac{1}{n_1}\sum_{i=1}^{n_1} Y_i \\ \frac{1}{n_2}\sum_{i=n_1+1}^{n} Y_i\end{pmatrix} \qquad Var(\mathbb{b}_{2\times 1}) = \left(\frac{1}{\sigma^2}(\mathbb{X}^T\mathbb{X})\right)^{-1} = \sigma^2\cdot\begin{pmatrix}\frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2}\end{pmatrix} \qquad \mathbb{b}_{2\times 1} = \begin{pmatrix}\overline{Y}_1 \\ \overline{Y}_2\end{pmatrix}$$

$$E\left[c^T\mathbb{b}\right] = c^T\beta \qquad \hat{\theta} = c^T\mathbb{b} \qquad Var(\hat{\theta}) = c^T(\mathbb{X}^T\mathbb{V}^{-1}\mathbb{X})^{-1}c$$

$$\hat{\theta} = \overline{Y}_1 - \overline{Y}_2 \qquad Var(\hat{\theta}) = (1,-1)\begin{pmatrix}\frac{1}{n_1} & 0 \\ 0 & \frac{1}{n_2}\end{pmatrix}\begin{pmatrix}1 \\ -1\end{pmatrix}\cdot\sigma^2 = \sigma^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$$

$$\frac{\overline{Y}_1 - \overline{Y}_2}{\sqrt{\sigma^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0;1) \longrightarrow \frac{\overline{Y}_1 - \overline{Y}_2}{\sqrt{s^2_{res}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t(n-2)$$

Comparing means of multiple datasets of normally distributed data

Example: Write down a design matrix of a linear model comparing 3 datasets.

- discuss uniqueness (for given purpose) of your design matrix
- consider a test statistic T from before, is there a way to use T to test whether all 3 means are the same

$$\mathbb{X}_{n\times 3} = \begin{pmatrix} \mathbb{1}_{n_1\times 1} & \mathbb{0}_{n_1\times 1} & \mathbb{0}_{n_1\times 1} \\ \mathbb{0}_{n_2\times 1} & \mathbb{1}_{n_2\times 1} & \mathbb{0}_{n_2\times 1} \\ \mathbb{0}_{n_3\times 1} & \mathbb{0}_{n_3\times 1} & \mathbb{1}_{n_3\times 1} \end{pmatrix} \qquad \mathbb{b}_{3\times 1} \to \begin{pmatrix} \overline{Y}_1 \\ \overline{Y}_2 \\ \overline{Y}_3 \end{pmatrix}$$

$$n_1 + n_2 + n_3 = n$$

$$\mathbb{X}^*_{n\times 3} = \begin{pmatrix} \mathbb{1}_{n\times 1} & \begin{matrix} \mathbb{1}_{n_1\times 1} & \mathbb{0}_{n_1\times 1} \\ \mathbb{0}_{n_2\times 1} & \mathbb{1}_{n_2\times 1} \\ \mathbb{0}_{n_3\times 1} & \mathbb{0}_{n_3\times 1} \end{matrix} \end{pmatrix} \qquad \overline{Y} = \frac{n_1\overline{Y}_1 + n_2\overline{Y}_2 + n_3\overline{Y}_3}{n_1+n_2+n_3}$$

F-distribution

Definition

Let X_1 be random variable with χ^2 distribution with k_1 degrees of freedom and X_2 be a random variable with χ^2 distribution with k_2 degrees of freedom. Let X_1 and X_2 be independent. Then following test statistic follows the Fisher-Snedecor probability distribution with k_1, k_2 degrees of freedom respectively.

$$F = \frac{X_1/k_1}{X_2/k_2}$$

Example: Show that the F statistic can be used to compare variances of 2 datasets.

Definition (Submodel)

Let M be a linear model under which $\mathbf{Y} \sim N(\mathbf{X}\alpha, \sigma^2\mathbf{I})$ and M_1 linear model under which $\mathbf{Y} \sim N(\mathbf{U}\beta, \sigma^2\mathbf{I})$. Let $\mathbf{X}$ be of a type $n \times k$, $\mathbf{U}$ of type $n \times k_1$, let rank $r(\mathbf{X}) > r(\mathbf{U})$ and each column of $\mathbf{U}$ can be expressed as a linear combination of columns in $\mathbf{X}$. Then we call M_1 a submodel of M.

Theorem

Let $\mathbf{a}, \mathbf{b}$ be estimates of α, β respectively and assume that M_1 is a correct model. Let $S_e = (\mathbf{Y} - \mathbf{Xa})^T(\mathbf{Y} - \mathbf{Xa})$ and let $S_A = (\mathbf{Xa} - \mathbf{Ub})^T(\mathbf{Xa} - \mathbf{Ub})$, then

$$F = \frac{S_A/(r(\mathbf{X}) - r(\mathbf{U}))}{S_e/(n - r(\mathbf{X}))} \sim F_{dist}((r(\mathbf{U}) - r(\mathbf{X})), (n - r(\mathbf{X})))$$

*AN*alysis *Of* *VA*riance (ANOVA)

Corollary

Since a model $\mathbf{U}_{n\times1} = \mathbf{1}_{n\times1}$ is a submodel of any parametrization of a model with more different means, using the F statistic to test submodel $\mathbf{1}_{n\times1}$ against the full model with design matrix $\mathbf{X}$ is testing $H_0 : \mu_1 = \mu_2 = ... = \mu_k$ against $H_1 : \exists i, j$ for which $\mu_i \neq \mu_j$.

Example: Derive non-matrix formulas for computing One-way ANOVA test, arrange all subsequent intermediate results into ANOVA table.

$$S_A = (\mathbb{X}a - \mathbb{U}b)^T(\mathbb{X}a - \mathbb{U}b) = n_1(\bar{Y}_1 - \bar{Y})^2 + n_2(\bar{Y}_2 - \bar{Y})^2 + n_3(\bar{Y}_3 - Y)^2 = \sum_{j=1}^{k} n_j(\bar{Y}_j - \bar{Y})^2 = S_A$$

$$df(S_A) = k - 1$$

$$\begin{array}{l}
n_1 \begin{cases} \bar{Y}_1 - \bar{Y} \\ \vdots \\ \bar{Y}_1 - \bar{Y} \end{cases} \\
n_2 \begin{cases} \bar{Y}_2 - \bar{Y} \\ \vdots \\ \bar{Y}_2 - \bar{Y} \end{cases} \\
n_3 \begin{cases} \bar{Y}_3 - \bar{Y} \\ \vdots \\ \bar{Y}_3 - \bar{Y} \end{cases}
\end{array}$$

$$S_e = (Y - \mathbb{X}a)^T(Y - \mathbb{X}a)$$

$$Y_i - \bar{Y}_j$$

$$S_e = S_T - S_A$$

$$S_T = S_A + S_e$$

$$S_T = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n} Y_i^2 - n\bar{Y}^2$$

ANOVA table

Source of Variation	Degrees of Freedom (df)	Sum of Squares (S)	Mean Square (MS)	F-value
Factor A	df_A	S_A	$\mathrm{MS}_A = \mathrm{S}_A/\mathrm{df}_A$	$\mathrm{F} = \mathrm{MS}_A/\mathrm{MS}_e$
Error	df_e	S_e	$\mathrm{MS}_e = \mathrm{S}_e/\mathrm{df}_e$	
Total	df_T	S_T		

Theorem

For sums of squares in One-way ANOVA the following equality holds.

$$S_T = S_A + S_e$$

Formulas

Let n be a total number of observations and n_j be a number of observation belonging to j^{th} group, then:

$$S_T = \sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(y_i^2) - n\bar{y}^2$$

$$S_A = \sum_{j=1}^{k} n_j(\bar{y}_j - \bar{y})^2$$

$$S_e = S_T - S_A$$